Applied Mechanics and Materials Vols. 513-517 (2014) pp 620-623 *Online: 2014-02-06*
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doi:10.4028/www.scientific.net/AMM.513-517.620

A Research of the Application of the RFID in the Library Management

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Keywords: Library; Automatic Recognition Technology; Data Analysis; Library management; Technology Upgrades

Abstract. As the application of the information network technology continuously extended, library management service platform is experiencing a shift to the information service system. In the library information management field, researchers are seeking a library identification system which is more convenient and comprehensive. Under the circumstance of the information technology conditions, Radio Frequency Identification (RFID) technology completes the automatic identification of the electronic identification tags--the two-dimensional code. The RFID technology is a contactless scanning recognition technology. And the core technology of RFID can be described as follows: using two-dimensional code as the filter identification tool according to backup data of the book management.

Introduction

The construction goal of the new library is to build open cultural facilities for our readers other than hug collections of books. The development tendency of library is to provide public cultural services which are open, free and equal. With the capacity and scale of the libraries expand continuously, traditional library management methods are unable to meet the daily working demand of efficiency. Therefore, library book management methods are in high need of technological research upgrades. Currently, RFID radio frequency identification technology has been widely applied in the intelligent library managements. According to the recent application in library, the RFID technology performs well in the accuracy and safety field of automatic recognition. With the help of the RFID technology, we're able to make a comprehensive record of the library collections which means the simplification of the management of library books and the improvement of the quality of service provided by our library.

More than 300 university libraries in America are equipped with RFID system, China's Shenzhen Library, Shanghai Changing Library, Jimei University Chengyi College Library, Children's Library, Xiamen, Shantou libraries have implemented RFID technology. Practice has proved that the use of RFID library system, you can improve the retrieval speed, help organize collections, greatly reducing administrator workload. Full of humanity, "supermarket" of rapid self-circulation book system, is most welcome. September 9, 2008, the introduction of radio frequency identification technology National Library two formal reception of readers, which marks the RFID technology has matured, we have reason to believe that with the continuous development of RFID technology and standards of sound, bringing decline in the cost of RFID over barcodes plus magnetic stripe obvious advantages, RFID library management system is bounded to usher in a comprehensive and rapid development.

1. Brief Introduction of the function of the Automatic Recognition System

RFID radio frequency identification technology (Radio Frequency Identification) is an effective form of electronic identification tag , identifying objects by two-dimensional code (electronic barcodes) for RF scanning , the computer repository information on the existing Find object to

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search select to borrow books on the case in the database records. This automatic identification system can communicate with objects using non-contact means quick scan information, has greatly improved the efficiency of the office library.

RFID is based on the specific use of the environment in which the use of different equipment components to form into, it works basically the same, is represented by a signal transmitter, signal receiver and scanning transmission equipment. RFID technology emerged in the last century , RFID electronic tag specific development through smart tags, smart cards, RF cards, scanners constitute an information transfer scanned data subject, an object with a non-contact scanning data under the conditions of the exchange of information and by the introduction of the RF signal to identify the computer to exchange information. RFID also has a light weight, small size, waterproof, anti-magnetic , magnetic effect excellent features, data storage can be encrypted form for storage , and can be implemented in a wide range of scanning batch information reading function. RFID technology has been in the current library management system with a wide range of applications.

The principle of RFID scan information do not like the traditional barcode recognition process in order to align the scanned bar code data information content, as long as the book is scanned in the RF range of the scan, the system is able to read the information in the data tag, which and books placement, orientation, etc. does not matter, so that the efficiency savings books greatly accelerated.

RFID system information reading response is very fast , in one second of time can batch information read 20 books , has changed the traditional book information scanning a book to read once again work process , ideal for reading books information take a batch operation , the biggest advantage is that RFID can read books in which the process of moving information .

RFID, small size and easy installation. RFID library management requirements can be made of any convenient shape of book binding , and even more advanced RFID technology can be embedded directly into the gutter of books within the books combined with the electronic card to be long-term , and will not affect the appearance of books reading and books appearance.

RFID can erase the information on the book or customize storage location for the library management in more variable, book version replacing electronic tag can be rewritten in the book the original Call, ISBN and other major book information, and the database can erase the data a new data backup, to facilitate replacement management library. RFID also has the function of intelligent security library , book identification process is the ability to enhance the anti-theft book , books electronic label according to the scanning to determine whether the book is entrained , which is in the library with a particularly widespread commercial application .

2. The Working Principle of the Automatic Recognition System

Library RFID system generally consists of three major parts, namely, electronic RF card , as the book labels, electronic RF card chip and magnetic circuit by the coupling components, the electronic tag in the magnetic signal transmitter can be identified in the scan process to achieve information exchange with the scanning device , the chip with information storage function, the information entered on the books and storing , radio frequency card has advantages of long life , in the book can be added internally via embedded processing life of its existence , while the RF card information can achieve real-time read and write capabilities , changes, and loan information on the books can do immediately to save . RF scanner, the scanner reads the book as an important tool for information, the scanner can be divided into two types of portable and stationary scanning equipment, scanner by the signal receiver and converter, composed in scanned books information after the data can be converted and transferred to a computer via a transmission antenna database records PLC through a computer network architecture can achieve network communication functions. The third is a data receiving antenna , an antenna for receiving and reading scanner data signal transmission , the signal collected by the antenna , and the information into the computer data storage system , the computer changes the book information rewriting data in the radio frequency card information also be transmitted through the antenna .

The RFID systems operating mode is used for analog transmission signal conversion , when the RF card electronic bookmarks to access the scanner operating range when the RF card chip can

sense the faint electromagnetic signals scanner , electronic chips bookmark function is activated and transmitting data to the feedback information the scanner , the scanner receives the electronic bookmarks feedback information back to the chip can immediately after the conversion processing of the data signal from the antenna to a computer database , a library book management integrated intelligent software for real-time release for background scanning signal recording and processing , treatment advice and then after the antenna back to the scanner. Scanner based on the information to judge the legitimacy and other equipment to make the appropriate linkage directive. RFID system is divided into three basic wage frequency bands commonly used low-frequency band is scanned 30KHZ-300KHZ, high frequency scanning band can reach 3MHZ-30MHZ, UHF scanning band can reach 300MHZ-3GHZ, library scanner working band frequency and 13.65MHZ being concentrated in 125KHZ/134KHZ .

3. The Application of the Automatic Recognition Technology in the Book Management Field

The management device for library books in innovation and reform, well-known equipment manufacturers have on library management aspects of RFID technology for specific intelligence applications, and accordingly launched RFID Smart PC database processing system. RFID reading technology using electronic bookmark information for library lending overview, according to the corresponding operating requirements, borrow books, return, renewals and other functions can be directly implemented.

The application of the RFID systems in the library can be divided into four different kinds according to the specific usage occasions:

a.Self-service borrow equipment in libraries: use RFID technology to scan and record book information, readers can search the library book information registration processing with the help of self-charging devices, intelligent self-borrow equipment can also accomplish the book-return, book-loss, renewals and other services, and provide readers with a convenient lending environment;

b.The RFID system can greatly improves the work efficiency of librarians: the RFID electromagnetic induction devices are capable of completing batch scanning of books information, hand-held scanner can be used to quickly find and check out whether the books are placed in the correct placement bring more convenience for library management work;

c.The RFID system can effectively integrate resources and release the burden of manpower in the management of library, a large reading room, just need a librarian for the last ten thousand books will be able to conduct a comprehensive management and control, freeing up more efficiency, for the reader, automatic loan system and RFID scanning system can effectively reduce the processing time library , eliminating the inconvenience of readers queuing library;

d.The brightest spot of RFID is the application of library security systems: the application of magnetic stripe bounded books and embedded electronic bookmarks, make it possible to distinguish whether books is borrowed from normal procedures, while RFID system can accurately scan the book information, and to borrow books with the database data for comparison, reducing library alarm system appears false positives, false negative phenomenon, to provide a safe Books effective protection.

Automatic Recognition Technology is of great significance in the library informatization construction. So far, the application of this new technology has greatly improved the efficiency of library management, the process of borrowing books. The quality of the public library services, of course, has been remarkable improved. And the Automatic Recognition Technology is one of the most important tools for the new kind library. And closer attention should be paid to this new tendency.

Acknowledgements

The research was supported by Fundamental Research Funds for the Central Universities No.3142013099 and Fundamental Research Funds for the Central Universities No. JD1203B V2012.

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Applied Science, Materials Science and Information Technologies in Industry
10.4028/www.scientific.net/AMM.513-517

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